

Business Case

E-Commerce Delivery Performance Analysis

Domain: E-Commerce | Role: Business Analyst | Dataset: Olist Brazilian E-Commerce Public Dataset

1. Project Overview

This project analyses e-commerce order, delivery, customer, and review data to understand delivery performance, identify where delays are concentrated, and assess whether delayed orders are associated with lower customer satisfaction.

2. Business Problem

The e-commerce operations team lacks clear visibility into where delivery delays are occurring and whether delayed orders are associated with lower customer satisfaction. This makes it difficult to identify which areas should be prioritised for improvement.

3. Business Objective

To measure delivery performance, identify geographic patterns in delivery delays, compare customer satisfaction for delayed and on-time orders, and recommend practical areas for further operational investigation.

4. Key Business Questions

1. What is the overall on-time and delayed-delivery rate?
2. Which customer states have the highest volume and/or rate of delivery delays?
3. Are delayed orders associated with lower customer review scores?
4. Which geographic areas should be prioritised for further investigation?

5. Scope

In Scope

- Overall delivery performance
- Geographic delivery performance by customer state
- Customer review scores
- Comparison of delayed vs on-time orders
- Prioritisation of areas for further investigation

Out of Scope

- Seller-level performance analysis
- Product-category performance analysis
- Logistics route optimisation
- Demand forecasting
- Machine learning
- Warehouse optimisation
- Pricing analysis

6. Expected Business Outcome

- Establish a clear baseline of delivery performance.
- Identify geographic areas that contribute substantially to delivery delays.
- Understand the observed relationship between delivery delays and customer satisfaction.
- Provide data-driven recommendations for operational investigation and monitoring.

7. Key KPIs

KPI	Definition / Purpose
Total Eligible Delivered Orders	Delivered orders with a valid customer delivery date.
Delay Rate	$\text{Delayed delivered orders} \div \text{eligible delivered orders} \times 100.$
On-Time Delivery Rate	$\text{On-time delivered orders} \div \text{eligible delivered orders} \times 100.$
Average Delay Days	Average number of days late among delayed delivered orders.
Average Review Score – Delayed	Average order-level review score for delayed orders.
Average Review Score – On Time	Average order-level review score for on-time orders.